diffpair_test_routed

Design Report

Rev 1 | 2026-06-22 | jlcpcb

kicad-tools 0.14.0

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	7 (5 SMD, 2 THT, 0 other)
Nets	26
Traces	1690 segments
Vias	188
Board Size	100.0 x 80.0 mm

Design Overview

Theory of Operation

Differential Pair Test Board

Multi-protocol HSDI regression testbench

Epic #2556 Phase 4L (issue #2658)

Communication Interfaces

Protocol	Signals
USB	USB2_D+, USB2_D-, VBUS_USB

Power Architecture

Power Rails: PWR_FLAG

Assembly Notes

4 fine-pitch components

- **Fine-pitch components**: 4 (U2, U1, U3, U4)

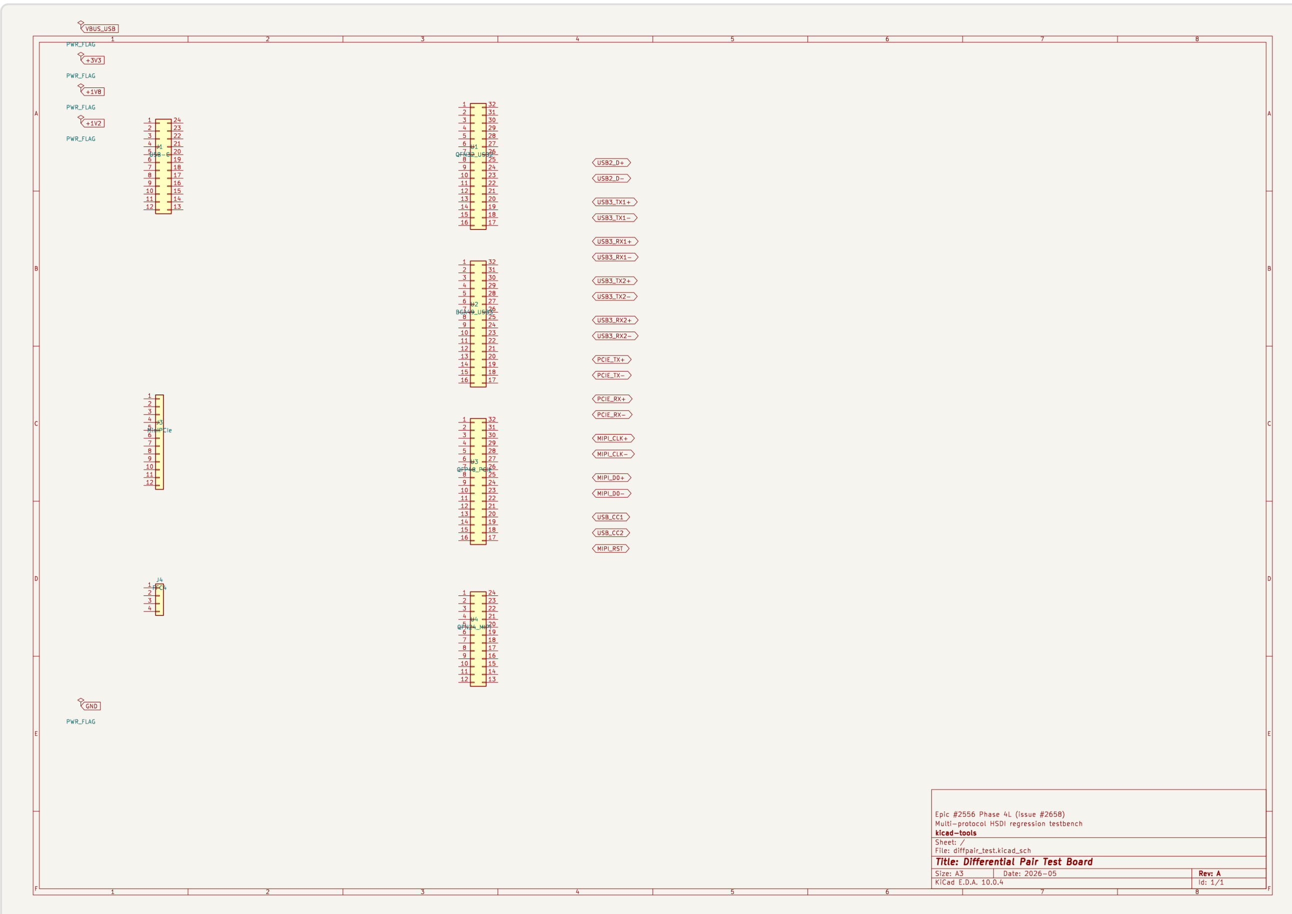
ERC Status

Metric	Count
Errors	0
Warnings	0

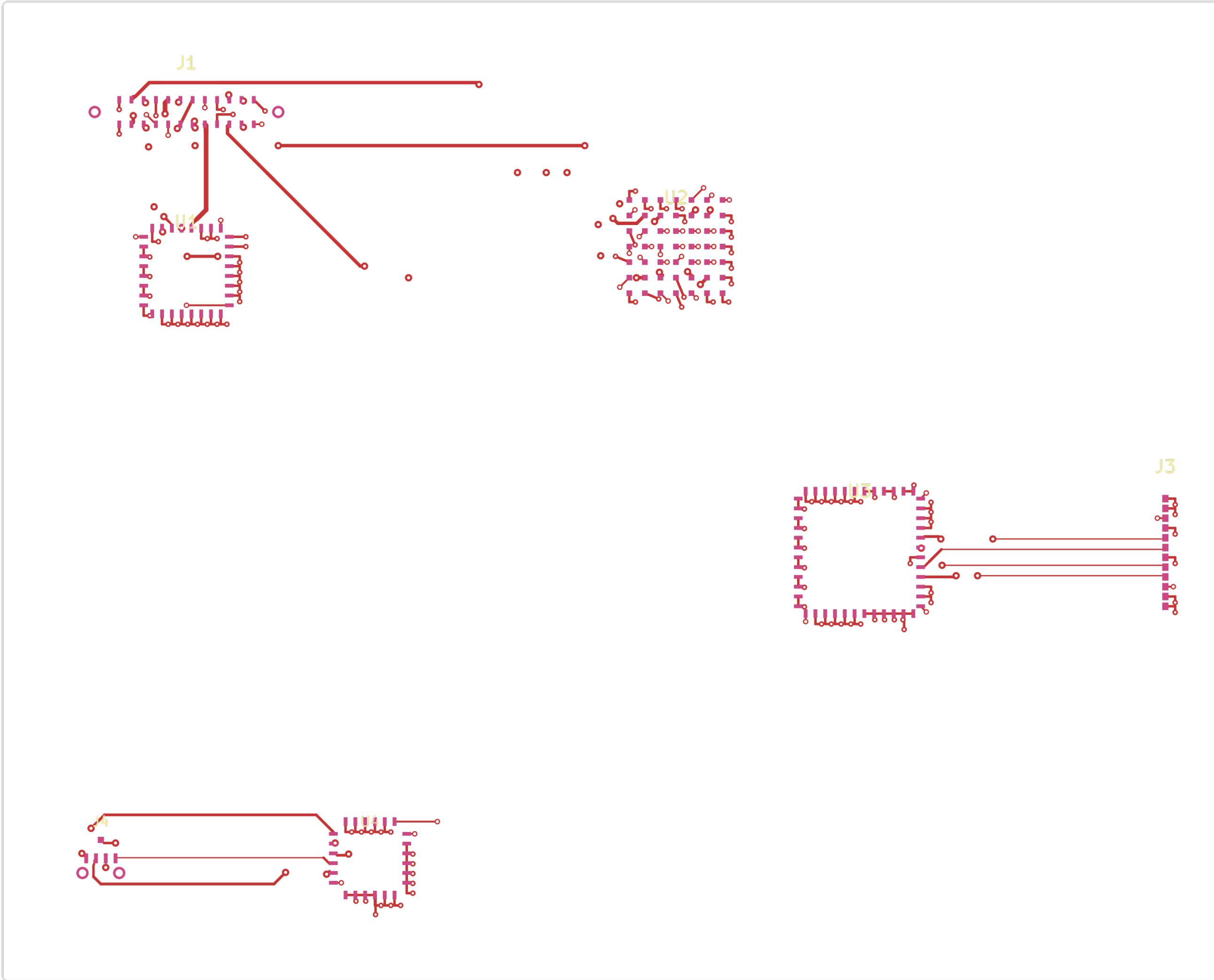
Status: SKIPPED -- ERC skipped by user request

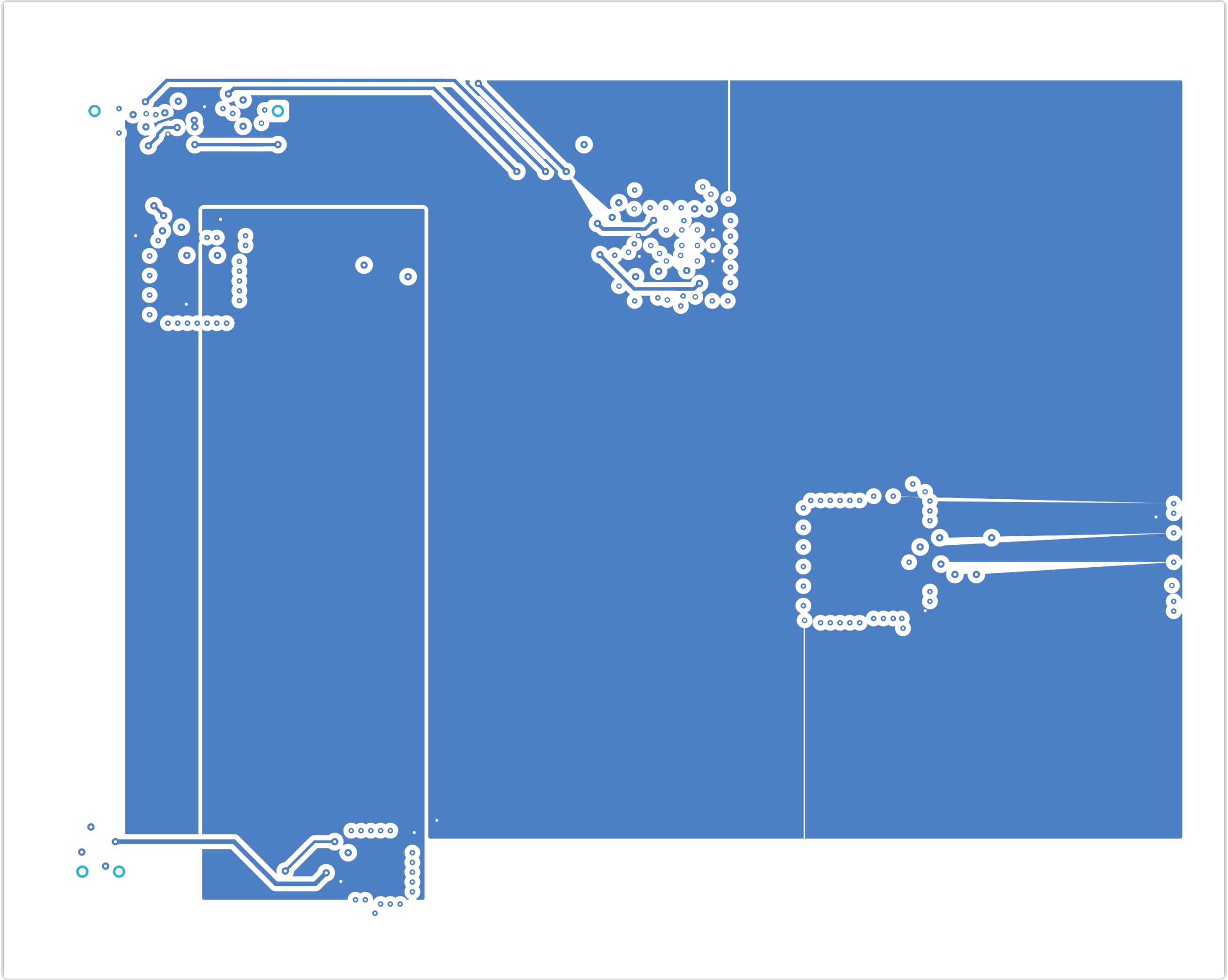
Schematic Overview

Schematic: diffpair_test

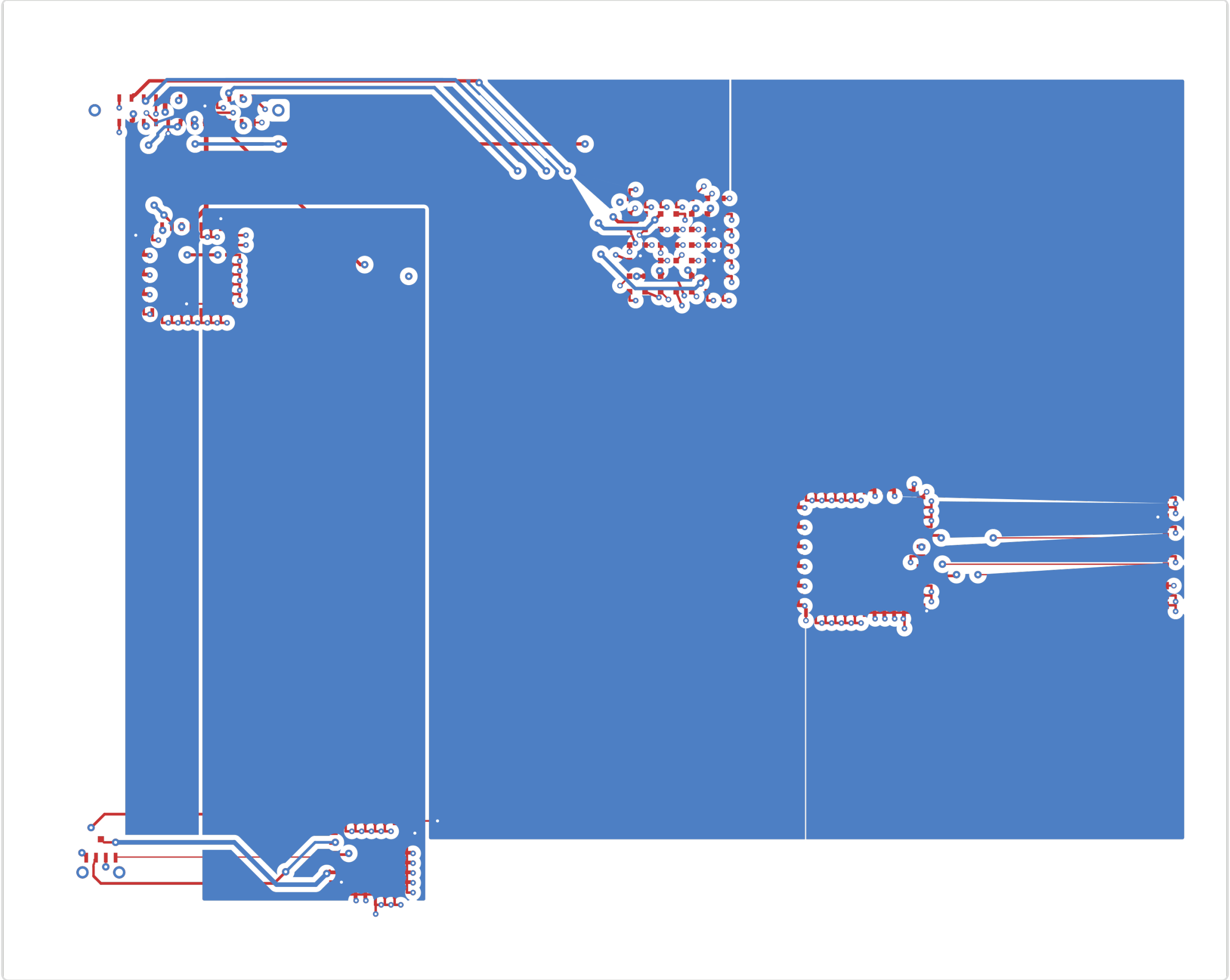


PCB Layout

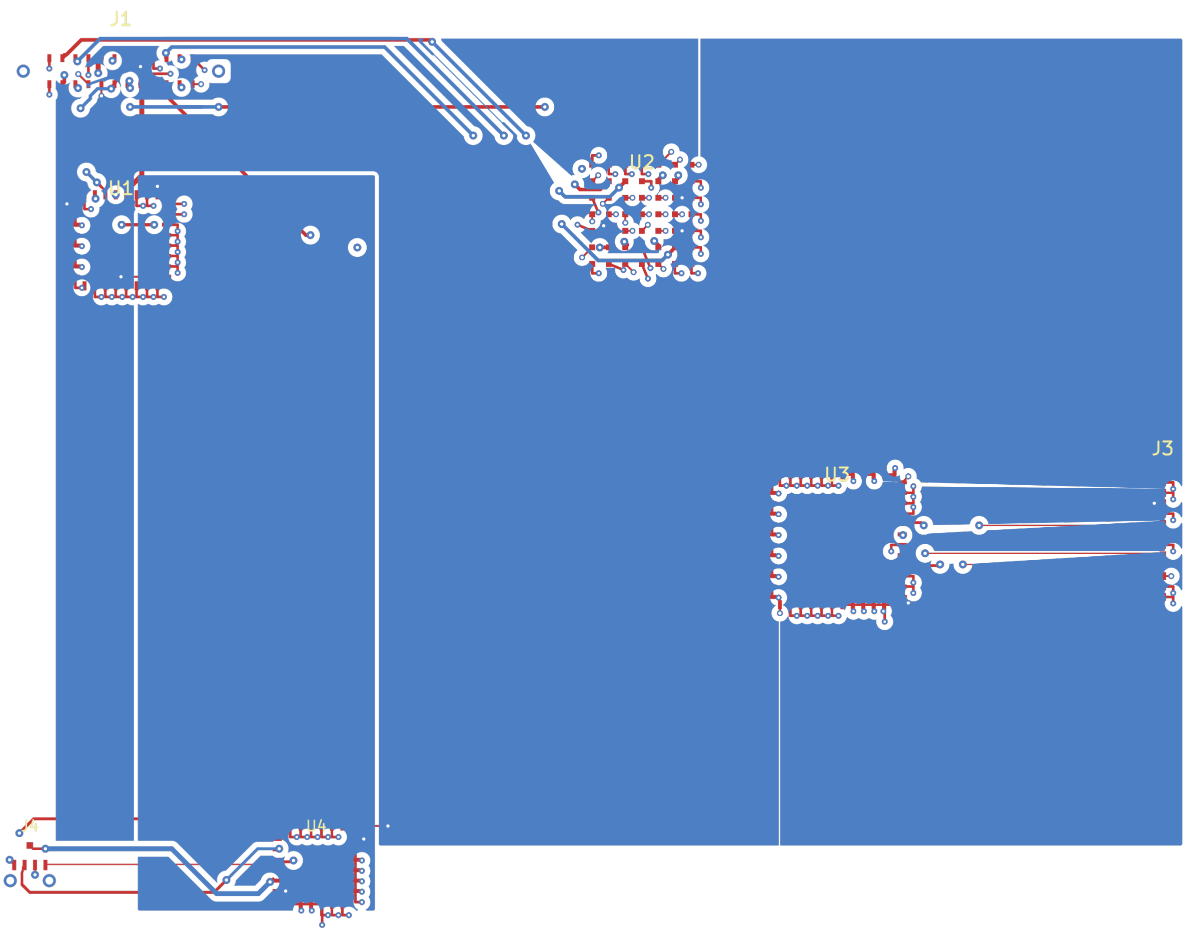




Copper

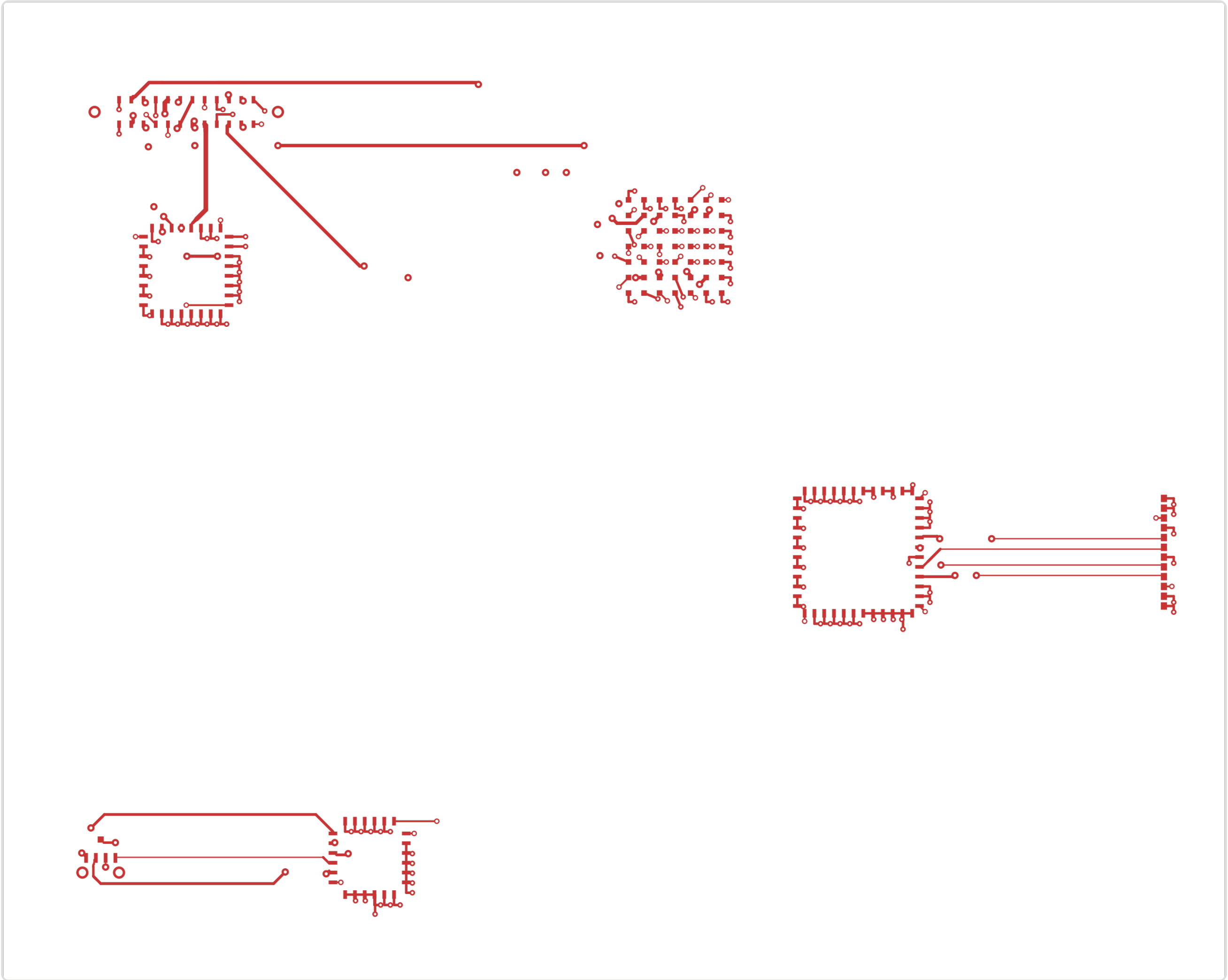


Assembly

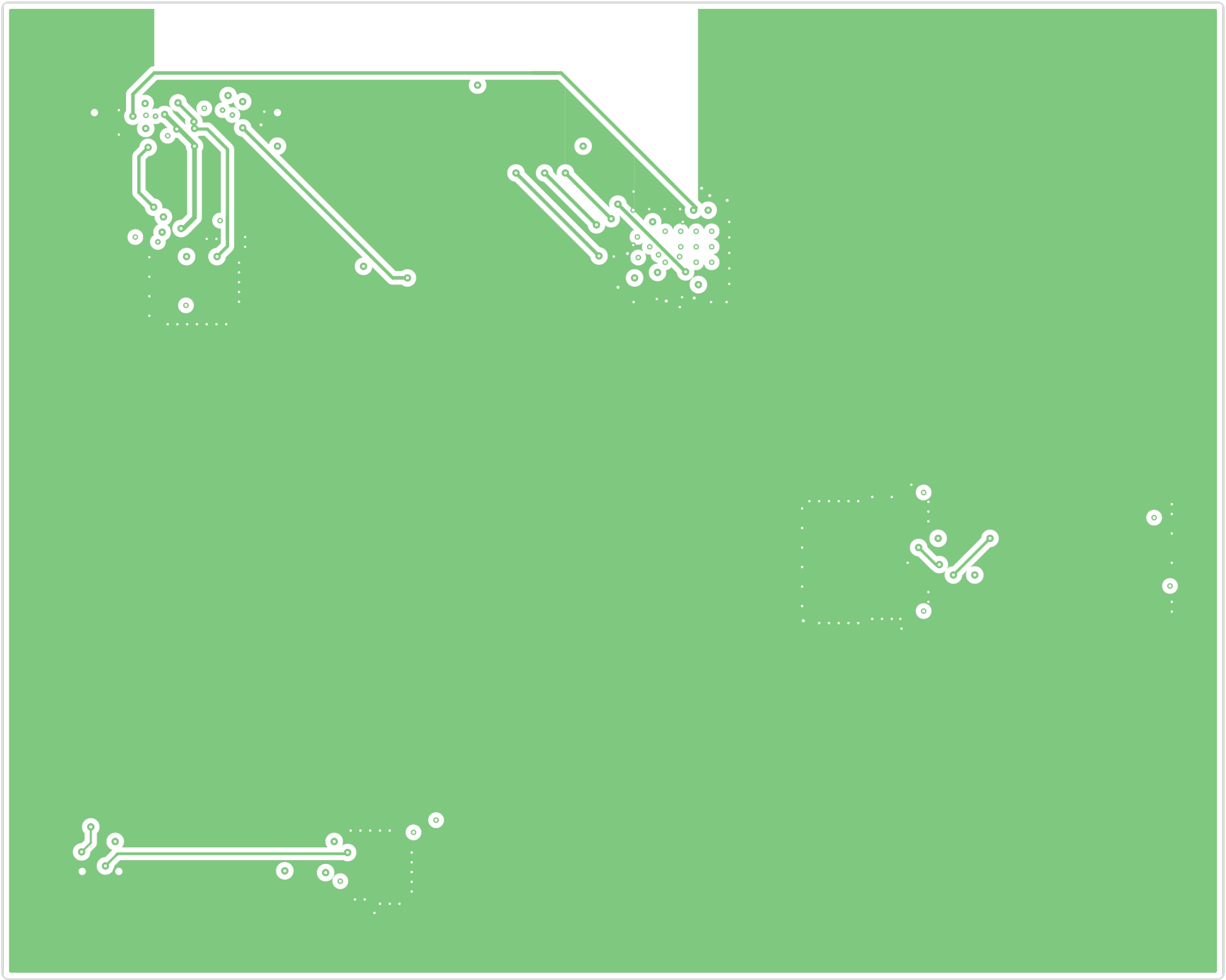


Copper Layers

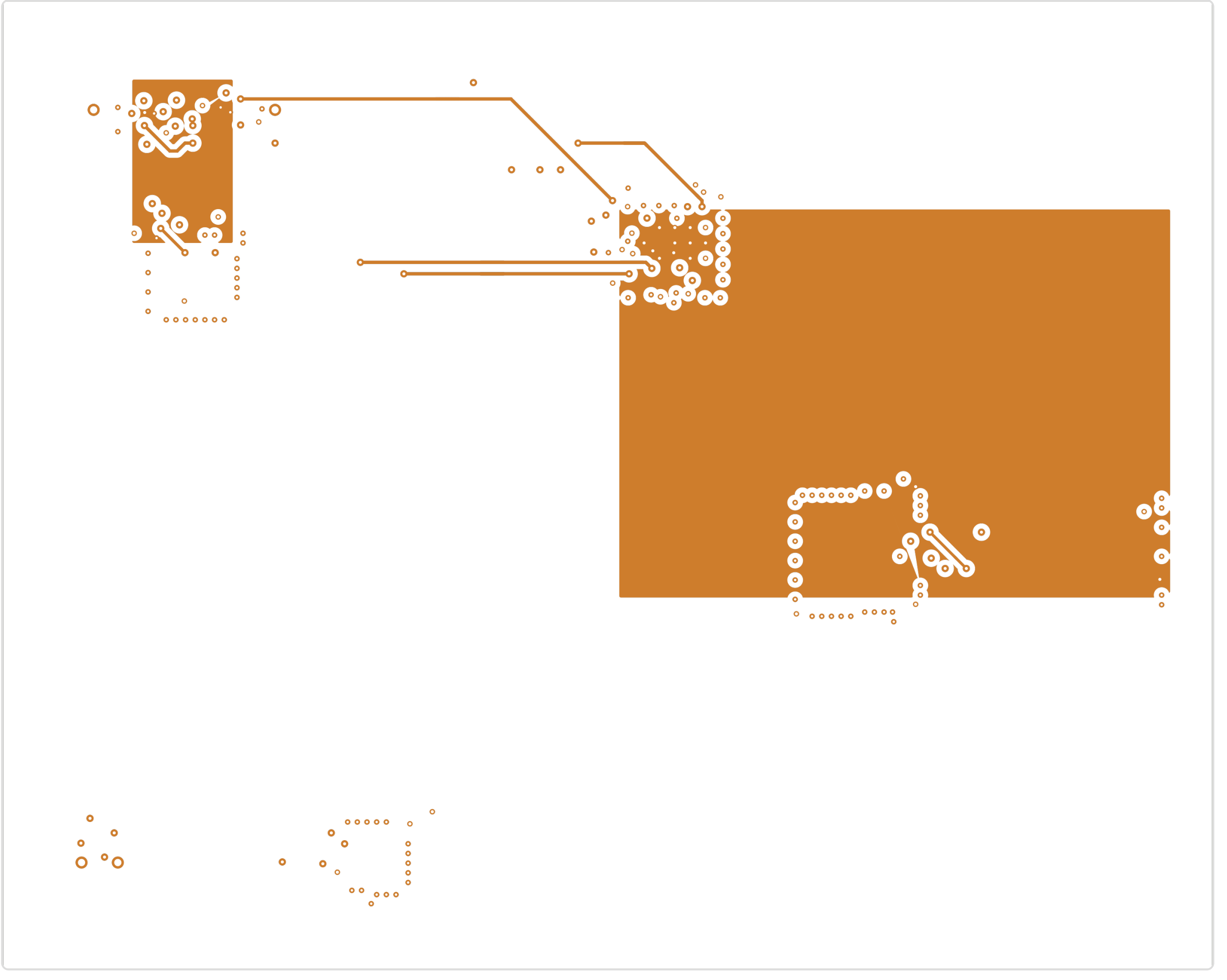
F.Cu



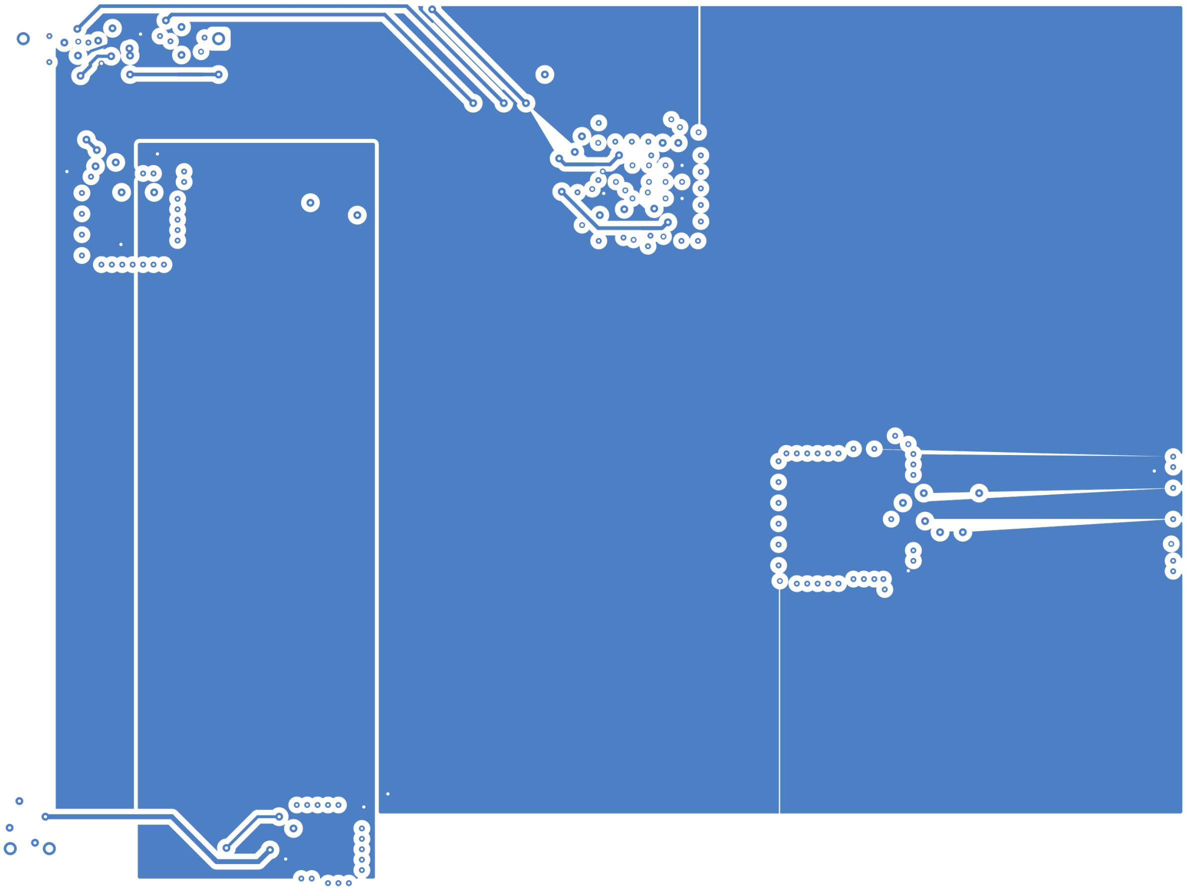
In1.Cu



In2.Cu



B.Cu



Bill of Materials

Value	Package	Qty	References
FFC4	FFC_4P_0.5mm	1	J4
MiniPCIe	PCIE_Mini_Edge	1	J3
USB-C	USB_C_Receptacle_USB2.0	1	J1
BGA49_USB3	BGA-49_5.0x5.0mm_Layout7x7_P0.5mm	1	U2
QFN24_MIPI	QFN-24-1EP_4x4mm_P0.5mm	1	U4
QFN32_USB2	QFN-32-1EP_5x5mm_P0.5mm	1	U1
QFP48_PCIe	LQFP-48_7x7mm_P0.5mm	1	U3

DRC Status

Metric	Count
Errors	10
Warnings	10
Blocking	10

Status: FAIL

Violations by Type

Violation Type	Count
silk_over_copper	4
clearance_segment_via	3
copper_sliver	3
silkscreen_text_height	2
kicad-cli:shorting_items	2
clearance_pad_via	1
connectivity	1
dimension_drill_clearance	1
silkscreen_over_pad	1
via_in_pad	1
kicad-cli:clearance	1
kicad-cli:hole_clearance	1

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- **[CRITICAL]** Fix 10 blocking DRC violations (clearance_segment_via (3), clearance_pad_via (1), dimension_drill_clearance (1); kicad-cli: shorting_items (2), clearance (1), hole_clearance (1))
- **[OPTIONAL]** Verify zone fill in KiCad for 5 zone-connected nets
- **[OPTIONAL]** Review 10 DRC warnings

Routing Status

Metric	Value
Signal Net Completion	100.0% (21/21)
Overall Completion	96.2%
Complete Nets	25 / 26
Zone-Connected Nets	5
Incomplete Nets	1
Unconnected Pads	14

Zone-Connected Nets

- +1V2
- +1V8
- +3V3
- GND
- VBUS_USB

Incomplete Nets

- GND

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~3.6 USD
Components (estimated)	~2.3 USD
Assembly (estimated)	~2.05 USD
Total (estimated)	**~7.95 USD**
Batch Quantity	5
Batch Total (estimated)	~39.74 USD